Bad Tails, Bad Times*

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Abstract

A lender's default loss is a put option on borrower revenue, so its value depends on the lower tail of the priced distribution, not on its variance. Because index options and corporate debt are priced by the same state prices, option-implied tail risk measures the terms on which lenders insure downside states, and it should matter for the real economy only when balance sheets are fragile enough that the downside state threatens default. We formalize this in a minimal model and test it on U.S. monthly data over 1996–2025, using a tail-risk factor extracted from the risk-neutral skewness and kurtosis of S&P 500 options. Tail risk predicts nothing unconditionally. In recession states it contracts production on impact, predicts substantially higher future volatility of real activity, and raises inflation through financing costs, while implied variance transmits mainly as disinflationary demand weakness. Conditioning on an observable leverage index attenuates the recession interactions while sharpening their precision, confirming fragility rather than the business cycle as the true state variable. The option surface is thus a real-time, forward-looking gauge of macroeconomic vulnerability that state-averaged frameworks largely miss.

JEL codes: E32, E37, E44, G12, G13, G17.
Keywords: option-implied moments, tail risk, balance-sheet fragility, leverage, external-finance premium, state-dependent local projections, structural VAR.

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1 Introduction

Option prices identify the risk-neutral distribution of returns on the underlying asset (Breeden and Litzenberger, 1978). The model-free methods of Britten-Jones and Neuberger (2000) and Bakshi et al. (2003) recover the moments of that distribution at any traded maturity. Macro-finance has made intensive use of the second moment. Implied variance and the variance risk premium forecast returns (Bollerslev et al., 2009, Drechsler and Yaron, 2011). They comove with monetary policy and risk appetite (Bekaert et al., 2013). They carry a term structure with distinct cyclical properties (Dew-Becker et al., 2017). The higher order moments of the same distribution have received sustained attention in asset pricing, from the repricing of index tail risk after the 1987 crash (Bates, 1991) to the pricing of tail-risk exposure in the cross section (Kelly and Jiang, 2014) and the cyclical behavior of ex-ante higher-moment risk (Gormsen and Jensen, 2025). Their content for real macroeconomic outcomes remains largely unexamined. This paper asks whether the higher order moments of the option-implied distribution contain information about future industrial production, unemployment, and inflation. It also asks when that information matters and through what economic mechanism.

The mechanism at the center of the paper is a simple credit-pricing argument. We state it here in plain terms because it disciplines everything that follows.

Consider a bank that lends to a firm. If the firm stays healthy, the bank is repaid in full. If the firm's revenue falls below the promised repayment, the bank absorbs the shortfall. The bank's loss therefore behaves exactly like a put option on the firm's revenue. It pays nothing in good times and grows as outcomes deteriorate below the repayment threshold. What matters for the value of this loss is how much probability sits below that threshold. That is a statement about the lower tail of the distribution. It is not a statement about the variance. Two revenue distributions can have identical means and identical variances and still imply very different losses for the bank, because one of them hides its bad outcomes in a fat lower tail. The higher order moments of a distribution, its skewness and its kurtosis, are exactly the summaries that describe this tail.

The next question is which distribution the bank uses to value the loss. A risk-averse lender does not value default losses at their expected frequency. It values them at state prices, which overweight the states in which wealth is scarce. State prices define the risk-neutral distribution. Our key assumption is that the same aggregate bad states drive both the stock market and firms' revenues, and that the same discount factor prices both index options and corporate debt. Under this assumption, the risk-neutral distribution recovered from S&P 500 options reveals the prices at which lenders value the aggregate component of

default losses. The higher order moments of that distribution are then direct and observable summaries of the lower-tail state prices that enter loan pricing.

The rest of the chain is mechanical. Competition forces the promised repayment to equal the risk-free repayment plus the risk-neutral expected default loss. The external-finance premium is proportional to that loss. When the priced lower tail deteriorates, expected default losses rise, promised repayments rise, and the premium widens. This happens even if option-implied variance has not moved, and even if the true probability of a bad state has not changed. Firms facing a higher premium buy fewer externally financed inputs, and production and employment fall in the next period. The full argument fits in one line.

priced tail shape $\longrightarrow$ risk-neutral default losses $\longrightarrow$ external-finance premium $\longrightarrow$ input demand $\longrightarrow$

The central feature of the mechanism is that transmission depends on balance sheets. In the model, the sensitivity of financing costs to priced tail risk depends on a single fragility statistic, the ratio of repayment obligations to downside pledgeable cash flow. When that ratio is below one, the loan is safe in every aggregate state and financing costs do not respond to priced tail risk at all. When it exceeds one, the downside state crosses the default boundary, financing costs respond, and the sensitivity rises with fragility. Recessions are times of elevated fragility, which is why crisis interactions matter empirically, but fragility is the primitive and the recession indicator is only its proxy. We therefore bring an observable fragility measure into every regression, the Chicago Fed's nonfinancial leverage index, and two findings confirm the reading. Conditioning on leverage attenuates the recession-interaction magnitudes in predictive regressions while sharpening their precision, exactly what one expects if the recession dummy partly proxies high-leverage states. And the local-projection results survive the leverage control essentially unchanged, indicating that the crisis-state transmission is not a mechanical artifact of omitted balance-sheet trends. This is the balance-sheet logic of the financial-accelerator literature (Bernanke and Gertler, 1989, Bernanke et al., 1999, Christiano et al., 2014), with priced tail risk as the impulse and leverage as the observable state.

A second feature is that the mechanism does not require option prices to forecast physical tail probabilities correctly. The risk-neutral tail probability rises either when the physical probability of a bad state rises or when investors attach a higher price to that state. The two components transmit differently. A higher physical probability weakens household demand through precaution and raises financing costs, so it contracts activity with an ambiguous inflation sign. A pure repricing of downside states leaves the physical income distribution unchanged, operates only through financing costs, and is therefore contractionary and un-

ambiguously inflationary when credit is exposed to the downside state. Both components contract activity. For this reason we treat the risk-neutral character of our measures as a feature rather than a defect. Financing terms are set by state prices and not by actuarial frequencies, so the risk-adjusted measure is the economically relevant one. Under the physical-news component, priced tail risk also predicts higher future volatility of activity, a prediction we test directly.

Our contribution is measurement rather than new theory. The accelerator mechanism is well established, and what is new is the input. The literature typically proxies priced default risk with credit spreads (Gilchrist and Zakrajšek, 2012) or estimates dispersion shocks inside a structural model (Christiano et al., 2014). We instead read the state prices of aggregate downside risk directly from the index-option surface. The reading is available at daily frequency, in real time, and moment by moment, so that the scale of priced risk and its higher order moments can be examined separately. The mechanism implies that the higher order moments should carry information beyond the variance, that the information should be concentrated in fragile states, and that an observable fragility control should absorb part of what crisis dummies capture. The data agree on all three counts.

Three sets of results organize the paper, and each is tied to an implication of the model.

First, transmission is state-dependent and runs through tail risk (Sections 4 and 6). In predictive regressions with the specification favored by the model, where macro variables and variance enter with one lag and tail risk with three, the tail factor is irrelevant unconditionally and strongly contractionary in recession states. The recession-state tail effect on industrial-production growth is -45.6 without a leverage control and -34.5 with it, with the t statistic rising from -3.6 to -5.2. In state-dependent local projections (Jordà, 2005, Auerbach and Gorodnichenko, 2012, Ramey and Zubairy, 2018) with wild-bootstrap inference (Mammen, 1993, Gonçalves and Kilian, 2004) and leverage controls, the normal-state responses are statistically indistinguishable from zero at every horizon while the crisis-state responses are large. A tail-risk shock lowers industrial-production growth by roughly 55 percentage points on impact, with the later sign reversal in the growth-rate projection reflecting the recovery of a depressed level, as the model's differencing logic implies. A volatility shock lowers production by 11 to 12 percentage points within a month and inflation by 0.9 percentage points at horizon three. Equality of the two states' coefficients is rejected with p values below 0.001 for the key channels. The same design applied to future realized macro volatility shows that tail-risk shocks predict substantially higher subsequent volatility of unemployment and production in crisis states, which is the physical-news prediction.

Second, the higher order moments share a dominant common tail component (Section 7).

In the model, downside skewness and excess kurtosis are increasing functions of the same rare-tail state once variance is held fixed, so their empirical principal component is a monotone normalization of the tail state. A six-variable structural VAR identified through heteroskedasticity (Rigobon, 2003, Lanne et al., 2010) confirms the prediction. The structural skewness and kurtosis shocks are nearly collinear. Skewness shocks account for a third of the forecast-error variance of kurtosis, and kurtosis shocks account for less than one percent of any macro variable. We therefore collapse the two higher order moments into a single tail-risk factor and conduct the analysis with two option shocks rather than three. The first principal component explains 82 percent of the variance of the two moments.

Third, the option block is predetermined and its state-averaged macro footprint is large (Section 7). Lagged macro variables contain no information about future option-implied moments, while lagged moments strongly predict macro outcomes. In a parsimonious five-variable structural VAR with leverage as an exogenous control, shocks to the option block account for 36 percent of the long-run variance of industrial-production growth and 18 percent of the long-run variance of inflation. Given the state-dependence established by the local projections, these figures are lower bounds on the importance of priced risk in fragile states. A companion volatility system, in which the macro block enters as realized variances, shows the same dilution, assigns the largest option-block share to the tail-risk channel into production volatility, and reveals that realized macro turbulence in turn feeds the priced tail.

Option-implied moments are conditional expectations under the risk-neutral measure, so some predictive content for subsequent outcomes is expected by construction, and we make no causal claim for the local projections. The substantive questions are the ones the design addresses. Do the moments contain information beyond standard predictors? They do, conditional on fragile states. Which moments matter for which outcomes? Tail risk dominates for real quantities and future real-side volatility, and the scale of priced risk matters most for inflation. Why does the information concentrate in bad states? Because financing costs respond to priced tail risk only when fragility places the downside state beyond the default boundary, and recessions proxy for exactly those states. The exercise inverts the customary direction of macro-finance inquiry. Rather than asking how macroeconomic conditions are priced in asset markets, we ask what asset markets measure for macroeconomics.

Section 2 places the paper in the literature. Section 3 describes the data. Section 4 reports the predictive regressions. Section 5 develops the model. Section 6 presents the state-dependent local projections. Section 7 presents the structural VAR evidence. Section 8 concludes.

2 Related Literature

Financial frictions and state-dependent amplification. In Bernanke and Gertler (1989) and Bernanke et al. (1999), external finance premia depend on expected default losses, generating amplification that is strongest when net worth is impaired; Christiano et al. (2014) estimate that shocks to the dispersion of idiosyncratic risk account for a substantial share of business-cycle variation. Gilchrist and Zakrajšek (2012) document the forecasting power of credit spreads and attribute it to the excess bond premium; Gilchrist et al. (2017) show that financial frictions shape firm-level price-setting in crises; López-Salido et al. (2017) show that boom-time credit-market sentiment predicts reversals in activity. Our model is a deliberately stripped-down member of this family: its contribution is to show that the lower-tail structure of default losses makes *option-implied higher order moments* the natural observable for the state prices these mechanisms transmit, to derive an explicit fragility statistic governing when the channel is active, and to connect that statistic to an observable leverage measure.

Uncertainty and the business cycle. Bloom (2009) establishes the recessionary effects of second-moment shocks through real-options channels; Jurado et al. (2015) construct econometric uncertainty indices; Basu and Bundick (2017) embed demand-side effects; Berger et al. (2020) attribute contractionary effects to realized volatility rather than forward-looking uncertainty news; Ludvigson et al. (2021) address the simultaneity of uncertainty and activity, the concern that motivates our identification strategy. This literature works almost uniformly with symmetric, second-moment notions of risk. The credit-pricing mechanism explains why symmetry is the wrong restriction precisely in bad states: default losses are lower partial moments, so the left tail — not the spread — is the payoff-relevant object. Evidence that risk asymmetry is itself cyclical (Segal et al., 2015, Salgado et al., 2019, Adrian et al., 2019) points in the same direction.

Option-implied information. Following Breeden and Litzenberger (1978), Britten-Jones and Neuberger (2000), and Bakshi et al. (2003), a large literature extracts conditional distributions from option panels. Bates (1991) reads crash expectations from index options; Bollerslev et al. (2009) and Drechsler and Yaron (2011) connect the variance risk premium to aggregate risk and risk aversion; Bekaert et al. (2013) decompose implied variance; Dew-Becker et al. (2017) price variance risk across the term structure. Kelly and Jiang (2014) and Giglio et al. (2016) construct tail-risk measures and evaluate them against macro tail outcomes. Closest to our question, Gormsen and Jensen (2025) document procyclical conditional tail risk in a multi-country option panel, and Bauer and Chernov (2024) relate

Treasury-implied skewness to biased beliefs and bond risk premia. We complement this work by mapping equity option-implied higher moments into real outcomes through an explicit credit-pricing channel.

Disaster risk and fragility. The rare-disasters literature (Rietz, 1988, Barro, 2006, Gabaix, 2012, Wachter, 2013) establishes that variation in the probability of large losses has first-order asset-pricing consequences; our aggregate-shock specification adopts its jump structure. On the origins of time variation in tail risk, the fragility-accumulation view — booms breed vulnerability — descends from Minsky (1986) and is formalized in Bordalo et al. (2018), Greenwood et al. (2022), and Brunnermeier and Sannikov (2014). Our fragility statistic is a balance-sheet object rather than a belief object, but the two views are complementary: leverage accumulated in booms is precisely what raises the statistic.

State-dependent transmission and identification. The state-dependent LP design follows Auerbach and Gorodnichenko (2012) and Ramey and Zubairy (2018), extending Jordà (2005); Plagborg-Møller and Wolf (2021) establish the LP–VAR equivalence that disciplines our joint use of the two methods. Structural identification through heteroskedasticity originates with Rigobon (2003), is developed for VARs by Lanne et al. (2010) and surveyed by Lütkepohl and Netšunajev (2017); inference uses the moving-block bootstrap of Brüggemann et al. (2016). Brunnermeier et al. (2021) apply the same identification philosophy to a ten-variable financial–macro system.

3 Data

3.1 Option-implied moments

We compute model-free risk-neutral variance, skewness, and excess kurtosis of one-month S&P 500 returns following Bakshi et al. (2003), using out-of-the-money calls and puts at the two maturities bracketing 30 calendar days and interpolating to the target horizon. Observations are sampled at month-end to align with the macroeconomic calendar. The sample is January 1996 through mid-2025.

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Data construction details: option data source and filters, strike-grid coverage, interpolation scheme, implementation of the moment formulas, and truncation adjustments.

Three measurement properties bear on the analysis. First, the moments are conditional expectations formed prior to the realization of the macro outcomes they may predict; Section 7 provides formal evidence of predeterminedness. Second, they are taken under the risk-neutral measure and therefore bundle physical probabilities with state prices. For the credit-pricing mechanism this is the relevant object: lenders price default losses at state prices, not actuarial frequencies. Third, risk-neutral skewness is negative throughout the sample (mean -1.9): index options price crash risk at all times (Bates, 1991); the empirical content lies in its variation.

3.2 The tail-risk factor

Anticipating the common-factor evidence of Section 7, we define

$$\text{tail_risk}_t \;=\; \text{PC}_1\big(\text{z}(-\text{skew}_t),\ \text{z}(\text{xkurt}_t)\big),$$

the first principal component of the standardized pair, signed so that higher values correspond to more downside risk. PC_1 explains 82.4% of the variance of the two-moment block, with equal loadings (0.707) and correlations of $+0.91$ with each input. Its correlation with implied volatility is -0.60. In the model, this factor is a monotone normalization of the risk-neutral downside probability, conditional on variance. We standardize the factor so responses are per one-standard-deviation shock.

3.3 The balance-sheet fragility proxy

The model's state variable is balance-sheet fragility, the ratio of repayment obligations to downside pledgeable cash flow. We proxy it with the Chicago Fed National Financial Conditions Index nonfinancial leverage subindex (FRED series NFCINONFINLEVERAGE), which summarizes household and nonfinancial-business leverage. The weekly series is averaged to monthly. Its correlation with the NBER indicator is 0.37 — leverage is elevated in recessions but not only in recessions, which is precisely why the model treats recessions as a proxy for fragility rather than as the primitive — and its correlation with the tail-risk factor is -0.10, so the fragility control is a state variable, not a competing shock measure. Leverage enters all regressions and local projections as a lagged control and the structural VAR as an exogenous regressor.

3.4 Macroeconomic variables

The macro block comprises industrial-production growth (annualized monthly log difference of INDPRO), the monthly change in the unemployment rate (UNRATE), and year-on-year CPI inflation (CPIAUCSL), from FRED, January 1996 through 2025. For the volatility exercises we construct the realized variance of each macro variable as the 12-month rolling variance of the monthly series, in logs. Crisis episodes are NBER recession months. The sample contains three recessions (2001, 2008–09, 2020); state-dependent results rest on these episodes, and the 2020 observations are extreme by any metric. We are explicit about the limitation where it binds.

3.5 Descriptive evidence

Figure 1 plots the option-implied moments against the macro variables with NBER recessions shaded. Three features motivate the formal analysis: all three moments respond to crisis episodes but with distinct signatures — variance spikes within recessions and reverts rapidly, whereas skewness and kurtosis exhibit pronounced movements surrounding stress episodes; the moments are far from collinear, with recurrent episodes of tail-pricing deterioration against quiescent variance — precisely the configuration in which the financing-cost channel of Section 5 moves with no signal from variance; and the macro series are dominated by a small number of extreme observations, which disciplines the inference standards we apply throughout.

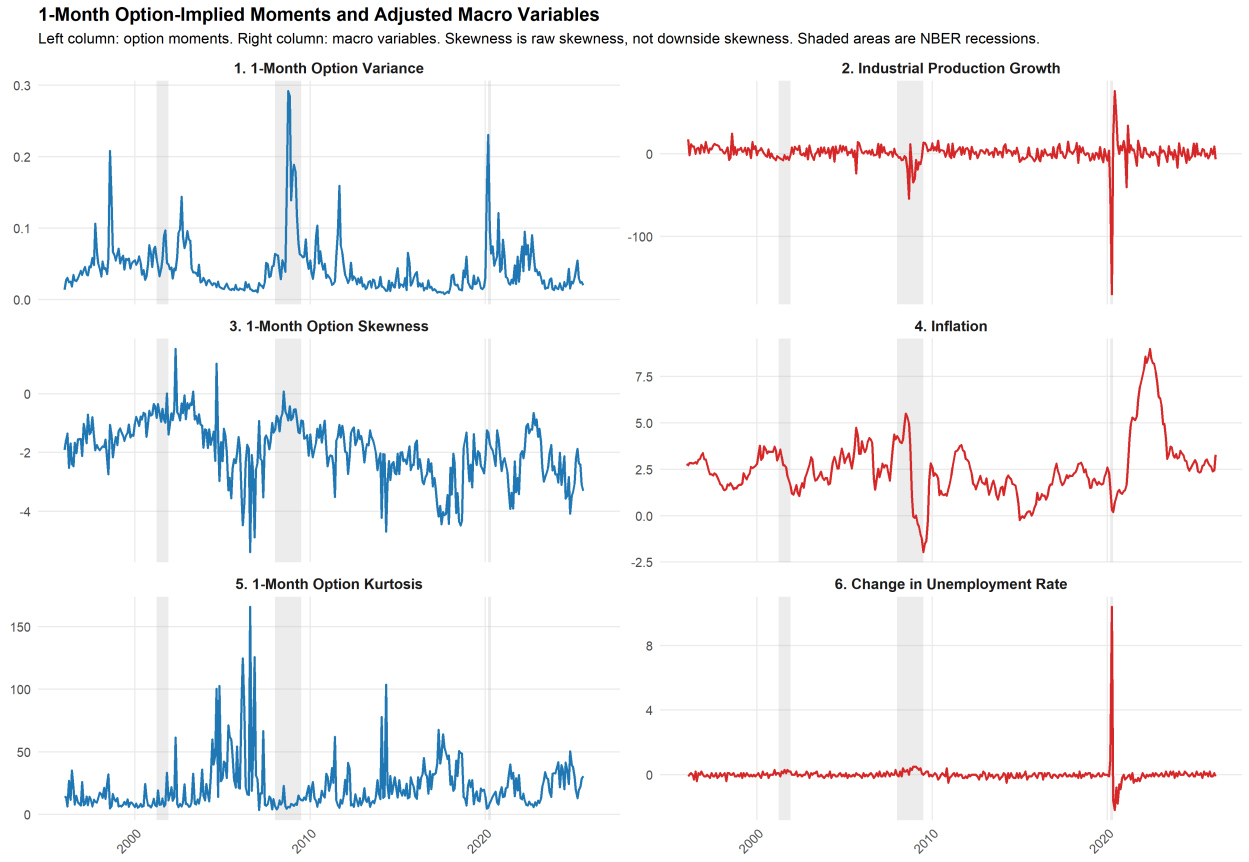


Figure 1: One-month option-implied moments (left column) and macro variables (right column), monthly, 1996–2025. Shaded areas are NBER recessions. Skewness is signed skewness.

4 Predictive Evidence

Before the structural work, we establish the reduced-form facts that any candidate mechanism must accommodate. The specification follows the timing the model suggests: macro controls and option-implied variance enter with one lag, while the tail-risk factor enters with three lags, reflecting its slower estimated pass-through. Inference uses Newey and West (1987) standard errors with four lags. Three models are reported for each outcome: Model 1 without crisis interactions, Model 2 adding NBER interactions, and Model 3 adding the leverage control. Reported tail and variance entries are recession totals (baseline plus interaction) in Models 2 and 3.

4.1 Conditional-mean regressions

Table 1: Predictive regressions for macro outcomes. Entries are recession-total coefficients on the option-implied measures (t statistics in parentheses; Model 1 reports unconditional coefficients). Macro controls and variance enter with one lag, tail risk with three. Newey–West standard errors (4 lags). *, **, *** denote 10, 5, and 1 percent significance.

	Model 1 (no crisis)		Model 2 (NBER)		Model 3 (NBER + leverage)	
Outcome	Tail	Variance	Tail	Variance	Tail	Variance
IP growth	-1.017	-4.340^{*}	-45.580^{***}	-2.355	-34.516^{***}	-2.873^{**}
	(-0.95)	(-1.79)	(-3.62)	(-1.64)	(-5.24)	(-2.06)
Δ Unemployment	0.063	0.195	2.480^{***}	0.076	1.746^{***}	0.111^{***}
	(1.19)	(1.38)	(3.86)	(1.49)	(4.48)	(3.08)
Inflation	0.011	-0.113^{**}	0.061	-0.315^{***}	0.265	-0.324^{***}
	(0.44)	(-2.13)	(0.41)	(-6.14)	(1.52)	(-6.94)

Three patterns organize Table 1. First, unconditionally (Model 1), the tail factor predicts nothing and variance is at best marginal. Second, in recession states (Model 2), tail risk becomes strongly contractionary for real activity — -45.6 for production growth, $+2.5$ for unemployment changes, both significant at the one percent level — while remaining irrelevant for inflation; variance is significantly disinflationary. Third, and central to the mechanism, conditioning on leverage (Model 3) *attenuates* the recession-total tail coefficients (production -45.6 to -34.5, unemployment 2.48 to 1.75) while *sharpening* their precision (t statistics rising from -3.6 to -5.2 and 3.9 to 4.5). This is exactly the pattern implied by the model's fragility statistic: the recession dummy partly proxies high-leverage states, so conditioning on leverage reallocates part of the measured amplification from the crisis interaction to the observable balance-sheet state and reduces residual noise. Variance effects on unemployment also become significant once leverage is held fixed.

4.2 Realized-variance regressions

If priced tail risk carries physical news about future bad states, it should forecast the future realized variance of macro outcomes, not only their means. We project the twelve-month-ahead realized variance of each macro variable on the same specification, controlling for lagged realized variance.

Table 2: Predictive regressions for 12-month-ahead realized macro variance. Format as in Table 1; lagged realized variances included as controls.

	Model 1 (no crisis)		Model 2 (NBER)		Model 3 (NBER + leverage)	
Future variance of	Tail	Variance	Tail	Variance	Tail	Variance
IP growth	23.5	60.0	2123.7***	112.9**	1977.0***	100.7***
	(0.71)	(0.80)	(4.34)	(2.00)	(5.26)	(3.10)
Δ Unemployment	0.090	0.137	6.401***	0.206*	5.682***	0.122**
	(0.88)	(0.60)	(4.00)	(1.69)	(6.16)	(2.25)
Inflation	−0.140	0.135*	0.328	−0.506**	1.843**	−0.382***
	(−1.18)	(1.91)	(0.29)	(−2.11)	(2.10)	(−3.12)

The volatility results mirror the means. Unconditionally there is no predictability. In recession states, tail risk strongly forecasts the future realized variance of production and unemployment, and the leverage control again attenuates the point estimates while raising precision. Inflation volatility is the exception: the recession-state tail effect appears only once leverage is held fixed, and we treat it cautiously below.

4.3 Quantile regressions

Following Koenker and Bassett (1978) and the growth-at-risk design of Adrian et al. (2019), we estimate quantile regressions of the macro outcomes on the option-implied moments. Point estimates in the lower tail carry the expected signs but are imprecisely estimated. Given three recessions, tail quantiles are identified from few effective observations; we read the quantile evidence as uninformative rather than contradictory, noting that Giglio et al. (2016) report analogous power limitations for systemic-risk measures in macro quantile forecasting.

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Table: Quantile regression estimates (10th, 25th, 50th, 75th, 90th percentiles) for the three macro variables.

4.4 Lag structure

A robust feature of the estimates is the differential timing: variance enters at a one-month lag, while the tail factor acquires significance at lags of approximately three months. All moments are measured at the one-month option horizon — three-month option moments perform strictly worse — so the long predictive lag of tail risk is not an artifact of the measurement horizon. The two-date model is deliberately silent on response horizons; a natural reading,

which we leave unmodeled, is that funding terms reprice gradually as existing contracts roll over, so the financing component of the tail channel accumulates over several months while the demand component of scale risk moves as soon as households revise spending.

5 A Minimal Mechanism Model

This section presents the model. It is a small, illustrative framework for interpreting the empirical responses of activity and inflation to option-implied variance and downside-tail risk. The option distribution is parameterized directly by total risk-neutral variance and a risk-neutral downside probability. Neither the observed inflation signs nor the relative strength of the channels is assumed; the model derives macroeconomic responses conditional on transparent common-state bridge assumptions, organizes impact signs, and does not attempt to identify the empirical innovations or match response magnitudes and horizons.

5.1 Environment, agents, and timeline

Time is discrete with two active dates. The public state at t includes household wealth, the firm's inherited external-financing need B_t, total option-implied variance V_t^Q, the physical downside probability p_t, and the relative state price of the downside state λ_t. The latter two jointly determine the risk-neutral downside probability q_t. The empiricist observes (V_t^Q, q_t) through the option surface but need not separately observe (p_t, λ_t). Three agents act. A prudent household chooses a spending commitment and saving. Competitive lenders finance B_t and quote promised repayment under the risk-neutral measure. A continuum of symmetric monopolistically competitive firms uses externally financed inputs, presets prices, and produces at $t+1$. Output, employment, and inflation are outcomes, not inputs into the credit block. The financing need B_t is inherited rather than chosen as a function of realized output, and the model is partial equilibrium.

5.2 Option-implied scale and tail risk

Let $D_{t+1} \in \{0, 1\}$ denote an aggregate downside state with physical probability p_t and define $\lambda_t \equiv M_t^D / M_t^N > 0$ from the conditional values of the stochastic discount factor. The risk-neutral downside probability is

$$q_t \;=\; \frac{p_t \lambda_t}{1 - p_t + p_t \lambda_t}, \qquad \frac{\partial q_t}{\partial p_t} > 0, \qquad \frac{\partial q_t}{\partial \lambda_t} > 0. \tag{1}$$

A deterioration in priced tail risk can reflect a higher physical probability, a higher state price, or both. Under Q, the index log return is specified so that V_t^Q is its exact risk-neutral variance and the jump is centered, giving the exact moments

$$-\text{Skew}_t^Q = \frac{q_t(1-q_t)(1-2q_t)J_r^3}{(V_t^Q)^{3/2}}, \qquad \text{ExKurt}_t^Q = \frac{q_t(1-q_t)\big[1-6q_t(1-q_t)\big]J_r^4}{(V_t^Q)^2}. \tag{2}$$

For $0 < q_t < q^\star \equiv (1-\sqrt{2/3})/2 \approx 0.092$, both expressions are strictly increasing in q_t when V_t^Q is held fixed. The empirical tail factor, a positive-weight combination of standardized $-$skewness and excess kurtosis, is therefore a monotone normalization of q_t conditional on variance; we use a positive tail-factor innovation and a positive q_t innovation interchangeably in the comparative statics.

5.3 Household demand

A prudent household ($u', v' > 0$, $u'', v'' < 0$, $v''' > 0$) chooses a spending commitment c_t for goods delivered at $t+1$. Future income loads on the same aggregate scale and downside states observed in options, $I_{t+1} = \bar{I} + \lambda_I \sqrt{V_t^Q}\,\xi_{t+1} - J_I D_{t+1}$, the model's common-scale bridge.

Lemma 1 (Precautionary demand)**.** *At a unique interior solution, higher V_t^Q lowers c_t, and higher p_t lowers c_t. A change in λ_t holding (V_t^Q, p_t) fixed leaves c_t unchanged.*

With aggregate demand $Y_{t+1} = \mathcal{D}_t P_{t+1}^{-\nu}$, $d_t \equiv \log \mathcal{D}_t$ increasing in c_t, the lemma yields the derived signs $d_V < 0$, $d_p < 0$, and $d_\lambda = 0$. The distinction between physical news and pure repricing matters below.

5.4 External-finance costs

The firm enters t with predetermined financing need $B_t > 0$. Pledgeable cash flow shares the two option states, $X_{t+1} = \bar{X} + \lambda_X \sqrt{V_t^Q}\,\eta_{t+1} - J_X D_{t+1}$. With the conditional put loss $\mathcal{P}(F; m, s) = E^Q[(F - m - s\eta_{t+1})^+]$, the risk-neutral expected default loss is

$$L^Q(F; V, q) = (1-q)\,\mathcal{P}\big(F; \bar{X}, \lambda_X\sqrt{V}\big) + q\,\mathcal{P}\big(F; \bar{X} - J_X, \lambda_X\sqrt{V}\big), \tag{3}$$

a lower partial moment of the cash-flow distribution. Convexity of the put payoff gives $L_q^Q > 0$ under nondegenerate downside-default exposure and $L_V^Q \geq 0$. Competitive lending, $R_f B_t = F_t - L^Q(F_t; V_t^Q, q_t)$, and implicit differentiation give, for the log funding wedge

$f_t \equiv \log(F_t/R_f B_t)$,

$$f_q \geq 0, \qquad f_V \geq 0, \qquad f_p = f_q \frac{\partial q}{\partial p} \geq 0, \qquad f_\lambda = f_q \frac{\partial q}{\partial \lambda} \geq 0, \tag{4}$$

strict under downside-default exposure and zero for a loan that is safe in every aggregate state.

5.5 Balance-sheet fragility

In the binary-cash-flow limit with normal-state cash flow $\bar{X}$ and downside cash flow $\underline{X}$, define balance-sheet fragility by

$$b_t \equiv \frac{R_f B_t}{\underline{X}}. \tag{5}$$

If $b_t \leq 1$, the loan is safe in both states, $f_t = 0$, and $f_q = 0$. If $b_t > 1$, default occurs only in the downside state, and zero profit yields the closed form

$$f(q, b) = \log\big(1 - q/b\big) - \log(1 - q), \qquad f_q = \frac{b-1}{(1-q)(b-q)} > 0, \qquad f_{qb} = \frac{1}{(b-q)^2} > 0. \tag{6}$$

Proposition 1 (Balance-sheet state dependence)**.** *Tail sensitivity of financing costs is zero when downside cash flow safely covers repayment ($b_t \leq 1$), becomes positive once the downside state crosses the default boundary ($b_t > 1$), and rises with fragility ($f_{qb} > 0$).*

Because b_t rises when external exposure increases or downside pledgeable cash flow falls, it is a parsimonious balance-sheet state. NBER recessions are interpreted empirically as a proxy for high-b_t states, not as a primitive of the theory, and the NFCI nonfinancial leverage index is our observable proxy for b_t itself. The attenuation-with-sharpening pattern of Table 1 is the empirical signature of this distinction.

5.6 Prices, output, and employment

A monopolistically competitive firm faces CES demand with elasticity $\epsilon > 1$ and produces with decreasing returns, $Y_{i,t+1} = A_{t+1} N_{i,t+1}^{\alpha}$, $0 < \alpha < 1$. The credit contract anchors a proportional funding wedge on externally financed inputs, with promised gross unit cost $\chi_t = w_t R_f \exp(f_t)$. The firm presets $P_{i,t+1}$ after observing demand and financing terms. The constant-elasticity first-order condition is a markup over marginal cost, and with $\theta \equiv$

$(1-\alpha)/\alpha > 0$ the symmetric equilibrium in logs is

$$\ell^*_{t+1} = K_{\ell,t} + \frac{f_t + \theta d_t}{1+\nu\theta}, \qquad y^*_{t+1} = K_{y,t} + \frac{d_t - \nu f_t}{1+\nu\theta}, \qquad n^*_{t+1} = \frac{1}{\alpha}\frac{d_t - \nu f_t}{1+\nu\theta} + K_{n,t}. \tag{7}$$

The inherited price is fixed when the option signal arrives, so the one-period inflation response equals the response of ℓ^*_{t+1}.

Proposition 2 (Transmission from primitive risk components)**.** *For $z \in \{V, p, \lambda\}$,*

$$y_z = \frac{d_z - \nu f_z}{1+\nu\theta} \leq 0, \qquad n_z = \frac{y_z}{\alpha} \leq 0, \tag{8}$$

strict for V and p, and strict for λ when credit has downside-default exposure. The inflation response is

$$\pi_z = \frac{f_z + \theta d_z}{1+\nu\theta}. \tag{9}$$

For scale and physical-tail innovations, inflation is negative, zero, or positive according as f_z is smaller than, equal to, or larger than $\theta|d_z|$. A pure downside-state-price innovation has $d_\lambda = 0$ and therefore $\pi_\lambda = f_\lambda/(1+\nu\theta) \geq 0$: pure repricing is contractionary and unambiguously inflationary when credit is exposed, and neutral for a safe loan.

Balance-sheet state dependence extends to both outcomes: within the fragile region, holding the demand component fixed across balance-sheet states,

$$\frac{\partial y_q}{\partial b} = -\frac{\nu f_{qb}}{1+\nu\theta} < 0, \qquad \frac{\partial \pi_q}{\partial b} = \frac{f_{qb}}{1+\nu\theta} > 0. \tag{10}$$

Fragility deepens the real contraction and shifts the price response upward. It does not force inflation to be positive, because the physical-demand component, when present, continues to push prices downward; and a strong balance sheet eliminates f_q but not the demand component, so the model does not force normal-state responses to be exactly zero.

5.7 Interpreting the empirical responses

A general empirical tail innovation contains both a physical-probability component and a state-price component; the model does not fix their relative contribution. Five readings connect the theory to the estimates.

1. *Both risk measures reduce activity.* Demand contractions and financing-cost increases lower activity; equation (8) gives the exact condition.

2. *Variance is disinflationary.* The estimated negative inflation response to variance is consistent with $\theta|d_V| > f_V$: precautionary demand dominates the financing-cost effect of scale risk.

3. *The tail-inflation response is theoretically two-sided.* If the observed tail innovation contains adverse physical news, financing pressure and demand weakness offset, and a near-zero or modest inflation response is consistent with $f_T \simeq \theta|d_T|$. A pure repricing movement is unambiguously inflationary when credit is exposed. Zero inflation does not mean zero transmission: under exact cancellation, $\pi_T = 0$ implies $y_T = d_T < 0$.

4. *Growth-rate reversals are expected.* The theory concerns initial level responses of output and employment. When a depressed level subsequently recovers, differencing mechanically produces opposite-signed growth and unemployment-change responses at later horizons; the two-date model does not generate or attempt to explain that recovery.

5. *Skewness and kurtosis share a factor.* Conditional on variance, both are increasing in the same q_t for $q_t < q^\star$ (equation 2); the empirical principal component is a monotone normalization of the tail state.

The model is silent about magnitudes, horizons, monetary policy, wage adjustment, and general-equilibrium feedback. It neither equates risk-neutral with physical probabilities nor treats local-projection innovations as structural.

6 State-Dependent Local Projections

Proposition 1 predicts that the transmission of priced tail risk into financing costs, and hence into real activity and inflation, is concentrated in fragile states. This section tests the prediction directly, estimating state-conditional impulse responses in the style of Jordà (2005), Auerbach and Gorodnichenko (2012), and Ramey and Zubairy (2018), following Plagborg-Møller and Wolf (2021) in reading the LP with lagged controls as identifying the same object as a VAR under the same conditioning set — with state-dependence entering transparently as an interaction.

6.1 Specification

For each macro variable $y_i \in \{\Delta U_t, \text{ip}_t, \pi_t\}$ and each standardized option-implied shock $x_{j,t} \in \{\text{vol}_t, \text{tail_risk}_t\}$, we estimate at each horizon $h \in \{0, 1, \ldots, 18\}$

$$y_{i,t+h} \;=\; \alpha_h \;+\; \delta_h\, S_t \;+\; \beta_h^N\,(1 - S_t)\, x_{j,t} \;+\; \beta_h^R\, S_t\, x_{j,t} \;+\; \boldsymbol{\gamma}_h'\, \boldsymbol{w}_t \;+\; u_{h,t+h}, \tag{11}$$

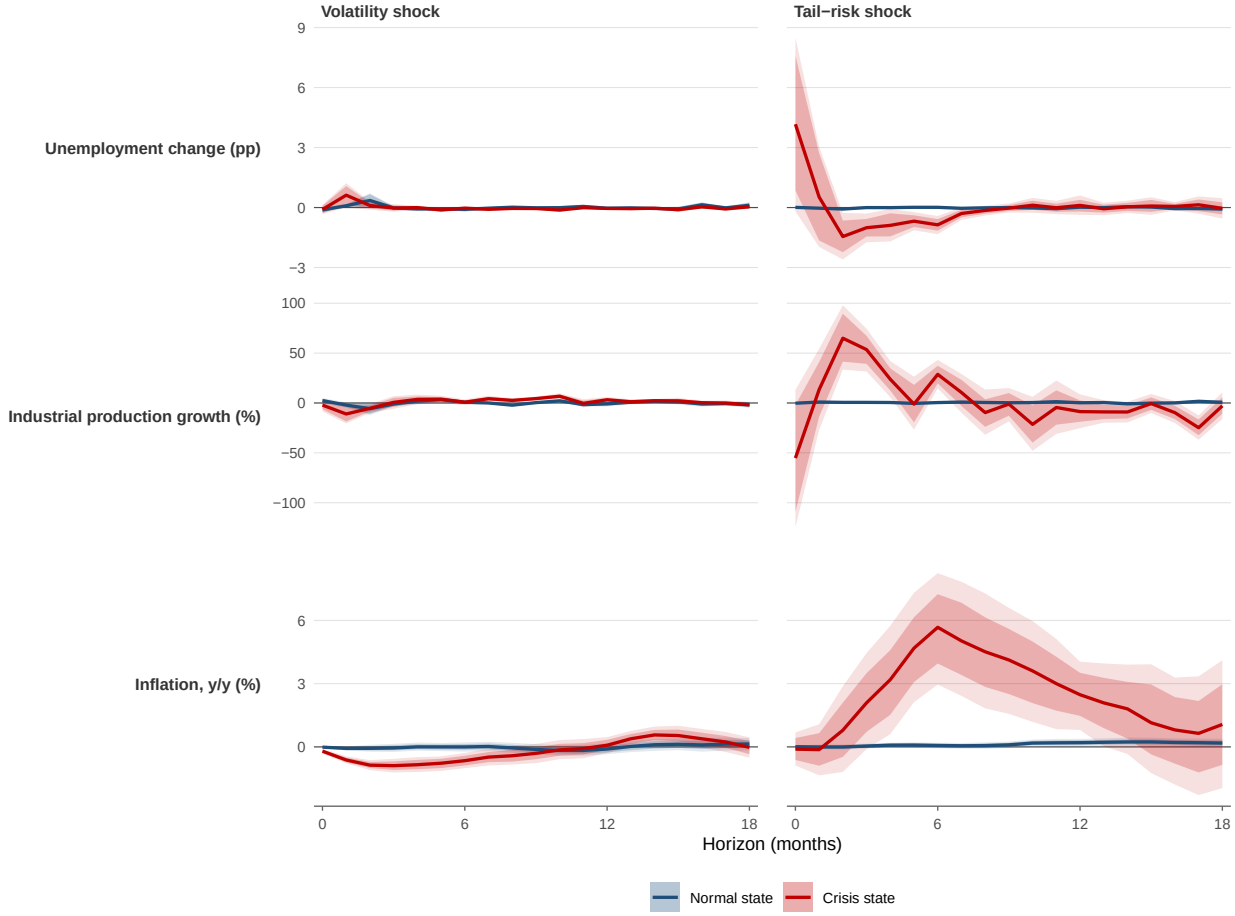
where S_t is the NBER recession indicator (*crisis state*, 28 of 354 sample months) and $\boldsymbol{w}_t$ collects two lags of the three macro variables, the two option-implied series, and the leverage index. The leverage lags control for the continuous balance-sheet state of Proposition 1, so the crisis interaction captures state-dependence beyond what observable fragility explains; results without the leverage control are essentially identical, indicating that the crisis-state transmission is not an artifact of omitted balance-sheet trends. Inference is by wild bootstrap with Rademacher weights (Mammen, 1993, Gonçalves and Kilian, 2004), 999 replications, regressors held fixed; bands are empirical quantiles, and the equality test $H_0 : \beta_h^N = \beta_h^R$ is computed from the bootstrap distribution of the difference. We also report an extended crisis window (NBER ± 3 months, 46 crisis months) that captures the periods of impaired balance sheets surrounding dated recessions. The local-projection innovations are not structural shocks of the model; the LP measures conditional co-movements given the lagged controls, and the model supplies their interpretation.

6.2 Results: conditional means

Figure 2 reports the state-dependent responses under the strict NBER definition. The normal-state responses are statistically indistinguishable from zero across shocks, horizons, and macro variables; all statistically detectable transmission is concentrated in the crisis state — the pattern Proposition 1 predicts, with the caveat that the model does not force the normal-state response to be exactly zero.

Figure 2. State−dependent local projections

Response to a +1 SD option−implied shock, controlling for leverage; wild−bootstrap 68% and 90% bands.



Notes: Crisis state = NBER recession months (28 of 354). Controls: two lags of system variables and of NFCI nonfinancial leverage (proxy for balance−sheet fragility b in the model). Wild bootstrap (Rademacher), 999 reps.

Figure 2: State-dependent local projections, NBER strict, with leverage controls.

Table 3: Peak crisis-state responses to a +1 SD option-implied shock, with leverage controls: the normal-state coefficient at the same horizon and the bootstrap p-value of $H_0 : \beta^N = \beta^R$. Panel A: NBER strict. Panel B: NBER ±3 months.

Response	Shock	Peak h	β^N	β^R	$p(\beta^N = \beta^R)$
Panel A. NBER strict (28 crisis months of 354).					
ΔU_t (pp)	vol	1	0.09	0.62	0.13
	tail-risk	0	0.01	4.17	0.13
Industrial production growth (% ann.)	vol	1	−2.10	−11.03	0.09
	tail-risk	2	0.62	64.94	**<0.001**
Inflation, y/y (%)	vol	3	−0.05	−0.89	**<0.001**
	tail-risk	6	0.07	5.67	**<0.001**
Panel B. NBER ±3 months (46 crisis months of 354).					
ΔU_t (pp)	vol	1	0.04	0.59	**0.02**
	tail-risk	4	−0.03	1.82	0.51
Industrial production growth (% ann.)	vol	1	−0.58	−12.15	**<0.001**
	tail-risk	6	0.26	35.41	**<0.001**
Inflation, y/y (%)	vol	3	−0.00	−0.83	**<0.001**
	tail-risk	18	0.15	3.22	**<0.001**

The tail-risk contraction is on impact, and the later reversal is the model's differencing artifact. In the crisis state, the response of industrial-production growth to a tail-risk shock is −55.2 pp at $h = 0$, turning positive (+64.9 at $h = 2$ under strict NBER) as the depressed level recovers. This is the pattern reading 4 of Section 5.7 anticipates: the theory speaks to the initial level contraction, and differencing mechanically produces the opposite-signed growth response during the recovery. The impact contraction aligns with the recession-total tail coefficient of −34.5 in the predictive regressions, which average the first months of the response under the three-lag timing. Under the extended window, the peak halves and shifts to $h = 6$, consistent with the strict-window peak being partly a 2020-specific timing artifact.

The volatility channel is sharp and survives the leverage control. A one-standard-deviation volatility shock lowers inflation by 0.9 pp at $h = 3$ and industrial-production growth by 11 to 12 pp within one month in the crisis state, against essentially zero in the normal state. The equality tests reject decisively for inflation under both crisis definitions ($p < 10^{-3}$) and for production under the extended window; under the strict window the production result is at the margin ($p = 0.09$). The volatility–unemployment channel appears under the extended window ($p = 0.02$). That the extended window — a closer proxy for the high-fragility region of Proposition 1 — sharpens rather than dilutes the results is itself consistent with the mechanism.

The inflation signs match the model's derived inequalities. Proposition 2 implies that a

volatility innovation is disinflationary when precautionary demand dominates ($\theta|d_V| > f_V$), which is what we estimate. The tail-risk inflation response is positive in crisis states (+5.7 pp at $h = 6$ under strict NBER, $p < 10^{-3}$), which in the model requires the financing component to dominate whatever physical-demand component the tail innovation carries — and is guaranteed for the pure repricing component, which is unambiguously inflationary when credit is exposed. The firm-level price-setting evidence of Gilchrist et al. (2017) supports the financing component directly: liquidity-constrained firms raised prices in 2008 while unconstrained firms cut them.

The normal-state responses are essentially null. This is the mechanical reconciliation with the state-averaged evidence of the next section: any linear model that constrains transmission to be regime-invariant estimates a convex combination of a large crisis-state coefficient and a null normal-state coefficient, with weights of order 92%–8% in favor of the normal state. Estimates that impose regime invariance are lower bounds on crisis-state transmission.

6.3 Results: macroeconomic volatility

Under the physical-news component of tail risk, priced tail risk predicts not only the level of future activity but its volatility. We re-estimate (11) with the dependent variable replaced by the log 12-month realized variance of each macro variable at horizon h, adding two lags of the dependent realized-variance series to the controls.

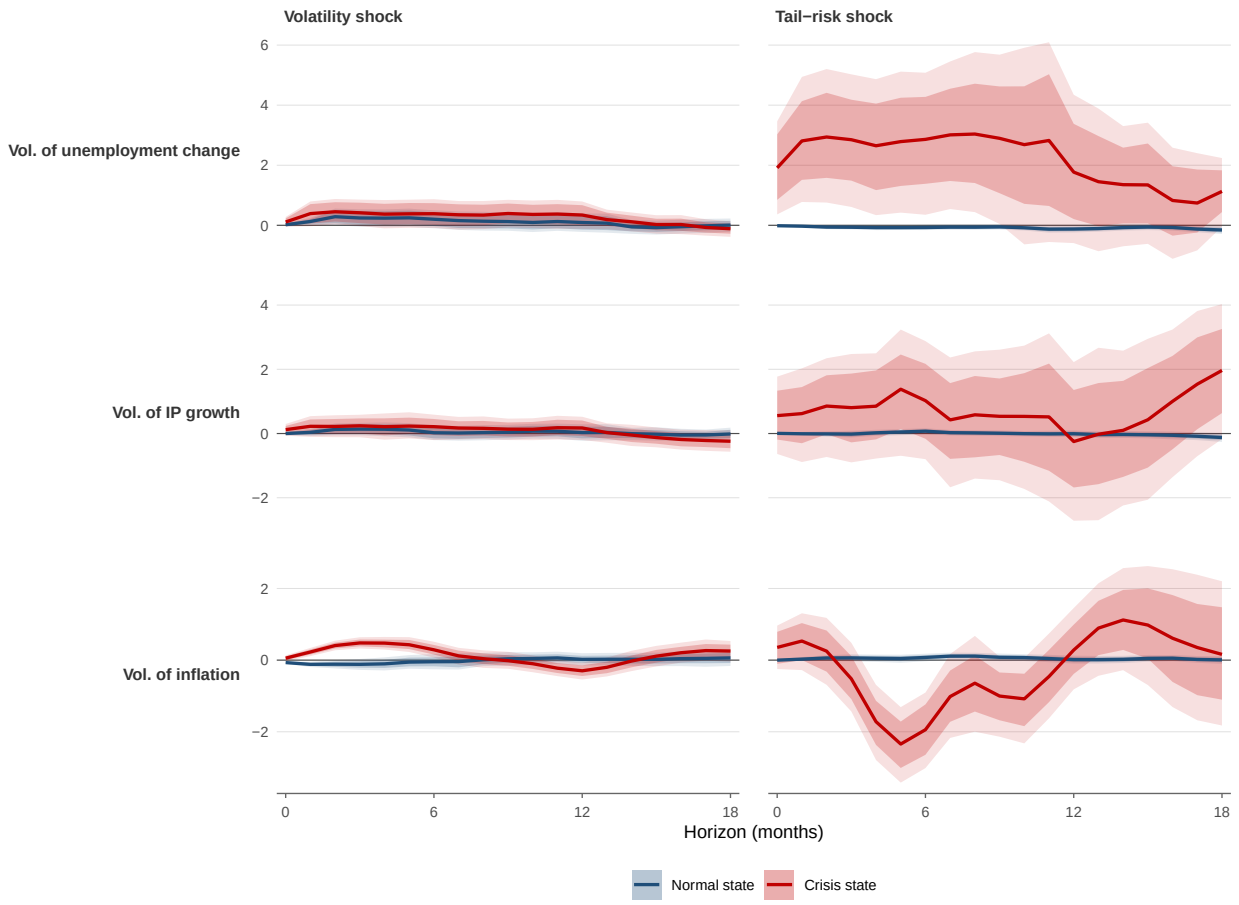


Figure 3: State-dependent local projections for macro volatility, NBER strict, with leverage controls.

Table 4: Peak crisis-state responses of log 12-month realized macro variance to a +1 SD option-implied shock, with leverage controls.

Volatility of	Shock	Peak h	β^N	β^R	$p(\beta^N = \beta^R)$
Panel A. NBER strict.					
ΔU_t	tail-risk	8	−0.06	3.04	**0.05**
ip_t	tail-risk	18	−0.12	1.97	0.10
π_t	vol	3	−0.12	0.48	**<0.001**
Panel B. NBER ±3 months.					
ΔU_t	tail-risk	8	−0.06	3.63	**<0.001**
ip_t	tail-risk	5	0.06	2.12	**<0.001**
π_t	vol	3	−0.15	0.40	**<0.001**

Three findings emerge. First, *tail-risk shocks predict substantially higher subsequent volatility of the real variables in crisis states*: the crisis-state response of unemployment-change volatility peaks at +3.0 to +3.6 log points around eight months ($p \leq 0.05$ under both crisis definitions), and the production-volatility response is significant under the extended window (+2.1 log points at $h = 5$, $p < 10^{-3}$). Normal-state responses are zero throughout. This is the LP counterpart of the realized-variance regressions of Table 2 and direct evidence for the physical-news component: tail innovations in crisis states carry information about the physical distribution, not only repricing. Second, volatility shocks predict modestly higher inflation volatility in crisis states. Third, the response of inflation volatility to tail-risk shocks is unstable across crisis definitions and we do not interpret it, noting only that the regression counterpart likewise appears only under the leverage control (Table 2).

6.4 Discussion

Three remarks discipline the interpretation. First, even under the extended window the sample contains only 46 crisis months, so crisis-state coefficients are imprecisely estimated — 90% wild-bootstrap bands routinely exceed the point estimates in absolute value. The results should be read as *qualitative*: they identify the states in which option-implied moments transmit, and the ranking of channels, more precisely than magnitudes. Second, the recession indicator and the leverage index are both proxies for the model's fragility statistic; firm-level measures of b_t — leverage, interest coverage, refinancing needs — map to Proposition 1 even more directly and are the natural next step. Third, the LP identifies each option-implied factor separately given lagged controls; the next section provides the complementary joint structural decomposition.

7 Structural VAR Evidence

The state-dependent LP delivers regime-conditional dynamics; it does not deliver a variance decomposition, a historical decomposition, or a joint orthogonalization of the option-implied moments. This section provides those complementary objects from a heteroskedasticity-identified structural VAR. Because the SVAR's reduced-form coefficients are common to normal and crisis states, its impulse-response and variance-share estimates are state-averaged and should be read as *lower bounds* on the crisis-state importance of the option block identified in Section 6. We proceed in two steps: a 6-variable specification testing the model's common-factor implication (Section 7.3), then the parsimonious 5-variable specification with

the tail-risk factor and an exogenous leverage control as our headline structural results (Section 7.4).

7.1 Specification and identification

We identify the impact matrix B in $\boldsymbol{u}_t = B\boldsymbol{\varepsilon}_t$ through heteroskedasticity (Rigobon, 2003, Lanne et al., 2010): with two volatility regimes in which $\mathrm{Var}(\boldsymbol{\varepsilon}_t)$ equals I_K and a diagonal Λ respectively, the pair $\Sigma_u^{(1)} = BB'$ and $\Sigma_u^{(2)} = B\Lambda B'$ point-identifies B up to sign and permutation provided the diagonal of Λ has K distinct entries. The maintained covariance break date is September 2008, with March 2020 as a robustness break; bootstrap Chow tests reject covariance constancy at both dates ($p = 0.01$ and $p < 0.01$). The multivariate ARCH-LM statistic rejects conditional homoskedasticity at any conventional level, confirming the presence of the identifying variation. Inference uses the moving-block bootstrap of Brüggemann et al. (2016) with block length 12 and 500 replications. Structural shocks are labeled by variance dominance. Both specifications use a VAR(2) with intercept (Hannan–Quinn selection; robust to $p = 3$); the headline 5-variable system includes two lags of the leverage index as exogenous regressors, so the structural shocks are orthogonalized net of the observable fragility state. We use the change in the unemployment rate because the level is borderline $I(1)$ in this sample, and excess kurtosis to remove the Gaussian benchmark. Cholesky-identified robustness runs are reported in the appendix.

7.2 Predeterminedness of the option block

Block Granger-causality tests on the 6-variable system discipline the direction of information flow:

H_0	F	df	p
Macro block does not Granger-cause option block	0.75	(27, 1992)	0.82
Option block does not Granger-cause macro block	2.69	(27, 1992)	6×10^{-6}

Lagged macro variables contain no incremental information about future option-implied moments; lagged moments strongly predict macro outcomes. This asymmetry is consistent with the timing of the model — option prices are set before outcomes realize and impound macroeconomic information before statistical agencies release it — and it supports the LP identifying assumption that option-implied moments are predetermined at monthly frequency.

7.3 The common-factor test

The model implies that, conditional on variance, downside skewness and excess kurtosis are increasing in the same tail state for $q_t < q^\star$ (Section 5.2), so the structural skewness and kurtosis shocks should be near-collinear. We test this in the 6-variable system $\boldsymbol{y}_t = (\text{vol}_t, \text{skew}_t, \text{xkurt}_t, \Delta U_t, \text{ip}_t, \pi_t)'$. The estimated regime-2 variance matrix is

$$\hat{\Lambda} = \text{diag}\big(1.51,\ 0.48,\ 0.094,\ 40.0,\ 3.61,\ 0.89\big),$$

with six distinct entries, so the label condition holds. The structural impact matrix loads the skewness shock on the kurtosis equation with coefficient -13.0 and the kurtosis shock with $+18.3$ — comparable magnitude, opposite sign. In the variance decomposition (Table 5), skewness shocks account for 32.9% of the forecast-error variance of kurtosis at $h = 24$, while kurtosis shocks account for less than 1% of every macro variable at every horizon. The two structural shocks are two windows onto a single latent factor, as the model predicts. The PCA reduction is therefore a structural conclusion, not a convenience; and consistent with the model's warning that the moments are not separately informative, re-estimating the predictive regressions with separate moments versus the tail-risk factor produces very similar results.

Table 5: Forecast-error variance shares (%) of the three option-implied moments at $h = 24$ in the 6-variable SVAR. Diagonal is own-shock share. The bold cell is the object of interest: skewness shocks explain a third of kurtosis variance.

Response	ε^{vol}	$\varepsilon^{\text{skew}}$	$\varepsilon^{\text{xkurt}}$
vol	40.5	1.7	3.9
skew	10.9	64.3	6.9
xkurt	3.6	**32.9**	56.2

7.4 Main structural results: the 5-variable specification

Replacing skewness and kurtosis with the tail-risk factor, and adding two lags of leverage as exogenous controls, gives $\boldsymbol{y}_t = (\text{vol}_t, \text{tail_risk}_t, \Delta U_t, \text{ip}_t, \pi_t)'$, monthly, April 1996 to August 2025 ($n = 352$). The estimated regime-2 variance matrix,

$$\hat{\Lambda} = \text{diag}\big(1.46,\ 0.41,\ 40.6,\ 3.75,\ 0.85\big),$$

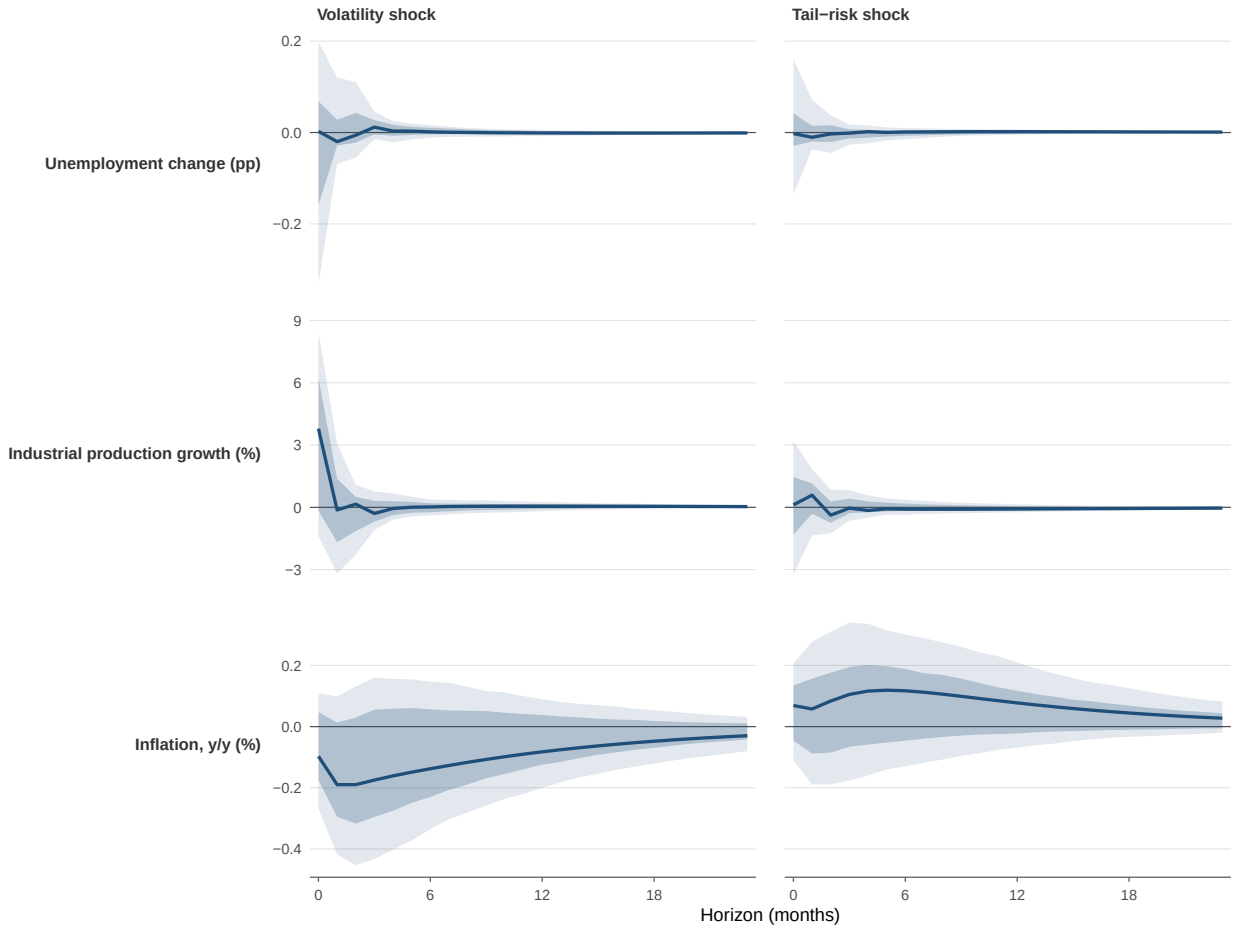
has five distinct entries; the label condition holds, and the cross-loadings of the two option shocks on each other are small — the collinearity of the 6-variable system is resolved.

The results are essentially identical without the leverage control, which is the structural counterpart of the LP robustness noted in Section 6.

Impulse responses. Figure 4 reports the responses of the three macro variables to the two structural option shocks with 68% and 90% moving-block bootstrap bands. Volatility shocks lower industrial-production growth by up to −1.5 pp within two months and inflation persistently to a trough of −0.2 pp at horizon three, with bands excluding zero; the tail-risk shock transmits mainly to inflation. The IRFs report the *average* transmission across states, and their magnitudes relative to the crisis-state LP estimates quantify how much the state-averaging dilutes: the crisis-state volatility response of production is roughly seven times the state-averaged structural response.

Figure 3. Structural VAR: option shocks and macro responses

Impulse responses to a +1 SD structural option shock; 68% and 90% moving−block bootstrap bands.



Notes: 5−variable SVAR with NFCI nonfinancial leverage (two lags) as an exogenous control. Identification through changes in volatility (id.cv, break 2008−09). 500 bootstrap replications, block length 12.

Figure 4: Structural VAR: option shocks and macro responses (5-variable specification, exogenous leverage).

Variance decomposition. Table 6 reports forecast-error variance shares at the 24-month horizon. Volatility shocks account for 34.6% of ip-growth variance and 11.5% of inflation variance; the tail-risk shock contributes a further 6.2% of inflation variance and small shares elsewhere. Interpreted as lower bounds on the crisis-state importance identified by the LP, the option block accounts for at least 36% of long-run ip-growth variance and at least 18% of long-run inflation variance.

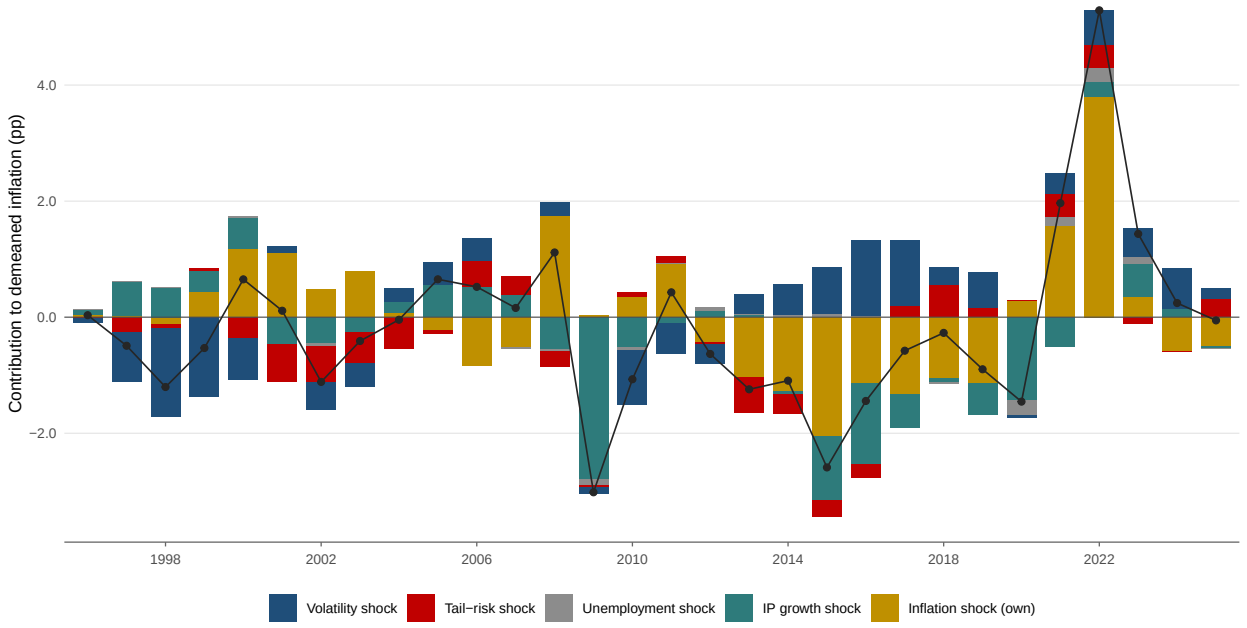
Table 6: Forecast-error variance shares (%) at $h = 24$ in the 5-variable SVAR with exogenous leverage. Rows sum to 100. Read as *state-averaged* shares; LP estimates imply these are lower bounds on crisis-state importance.

Response	ε^{vol}	$\varepsilon^{\text{tail}}$	$\varepsilon^{\Delta U}$	ε^{ip}	ε^{π}
ΔU_t	2.9	0.8	70.8	9.7	15.8
ip_t	34.6	1.5	9.5	46.4	7.9
π_t	11.5	6.2	0.0	7.7	74.5
Option block (vol + tail-risk) total:			*ΔU: 3.7%*	*ip: 36.1%*	*π: 17.7%*

Historical decomposition. Figure 5 decomposes the demeaned path of year-on-year CPI inflation into the annual-average cumulative contributions of the five structural shocks. The 2021–2022 inflation episode is attributed predominantly to the own inflation shock but with a visible contribution from the accumulated volatility shock; the tail-risk contribution is smaller on average but positive in the post-2021 surge.

Figure 4. Historical decomposition of inflation

Annual−average structural−shock contributions; black line: model−implied demeaned inflation.



Notes: 5−variable SVAR with exogenous leverage control. Annual averages of monthly cumulative contributions.

Figure 5: Historical decomposition of inflation (5-variable specification, exogenous leverage).

7.5 The volatility system

The volatility LP of Section 6.3 has a structural counterpart. We re-estimate the SVAR with the macro block replaced by realized variances, $\boldsymbol{y}_t = (\text{vol}_t, \text{tail_risk}_t, \text{lrv}_{\Delta U,t}, \text{lrv}_{ip,t}, \text{lrv}_{\pi,t})'$, where lrv denotes the log 12-month rolling realized variance, with the same identification (VAR(2), covariance break at September 2008). The estimated regime-2 variance matrix $\hat{\Lambda} = \text{diag}(1.75, 0.38, 8.46, 0.56, 0.99)$ has five distinct entries, though the near-unit entry for inflation volatility indicates weak identification of that equation.

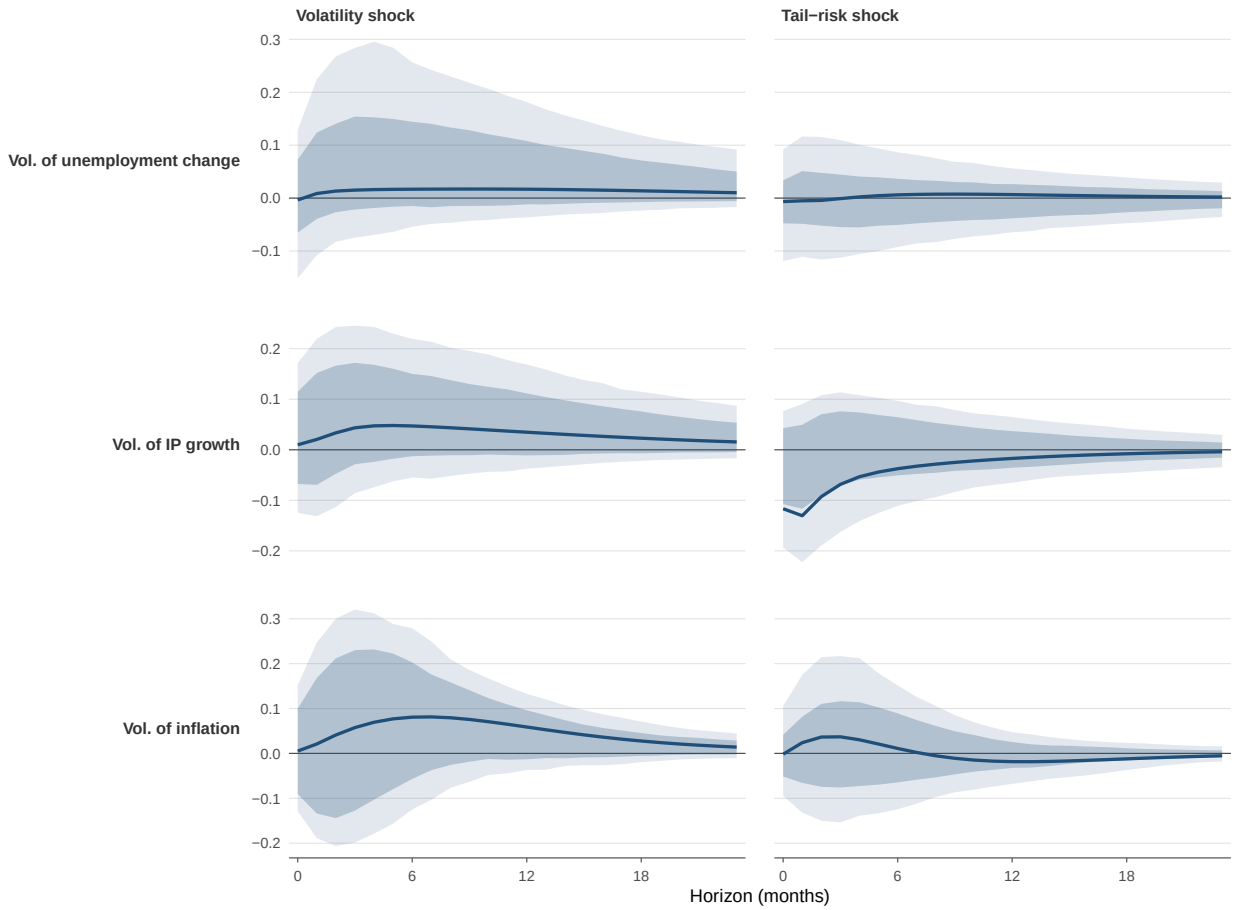
Two features distinguish this system from the level system, and both are informative. First, predeterminedness fails by construction: block Granger tests reject in *both* directions ($p = 2 \times 10^{-4}$ for macro-volatility $\to$ options, $p = 4 \times 10^{-3}$ for options $\to$ macro-volatility). This is expected — rolling realized variance is a backward-looking window of past shocks — and its converse is economically interesting: realized macro turbulence feeds tail pricing, with IP-volatility shocks accounting for 20% of the forecast-error variance of the tail-risk factor at $h = 24$. Markets appear to update the priced tail in response to realized macro volatility, consistent with the tail state being persistent and partially learned from data.

Second, the state-averaged impulse responses of macro volatility to option shocks (Fig-

ure 6) are small, with bands generally covering zero — precisely the dilution the LP implies when crisis-only transmission is averaged over a sample that is 92% normal months. The variance decomposition tells the same story at the margin: the option block accounts for 2.0% of unemployment-volatility variance, 10.2% of IP-growth-volatility variance, and 5.2% of inflation-volatility variance at $h = 24$, with the largest single option contribution being the tail-risk share of IP-growth volatility (6.9%) — the state-averaged image of the LP's strongest volatility result. The historical decomposition of unemployment volatility (Figure 7) attributes the 2020–2021 spike almost entirely to the own shock, with option-shock contributions visible mainly around 2008–2011. Taken together, the level and volatility systems deliver a consistent message: state-averaged structural objects bound the crisis-state importance of priced risk from below, and the state-dependence itself is carried by the local projections.

Figure 6. Volatility−system SVAR: option shocks and macro volatility

Responses of log 12−month realized macro variance to a +1 SD structural option shock; shaded areas: 68% and 90% moving−block bootstrap bands.



Notes: 5−variable SVAR (vol, tail−risk, three log realized variances), monthly. Identification through changes in volatility (id.cv, break at 2008−09). 500 bootstrap replications, block length 12.

Figure 6: Volatility-system SVAR: responses of realized macro variance to option shocks.

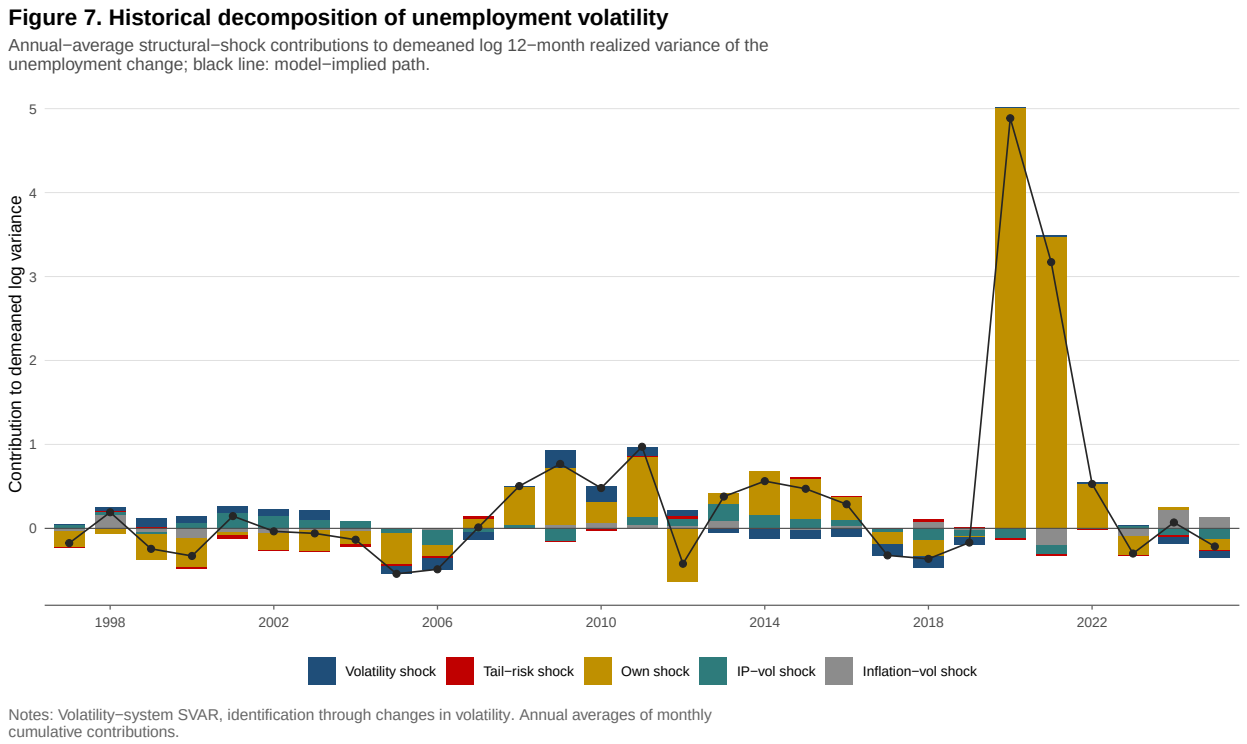


Figure 7: Historical decomposition of unemployment volatility (volatility-system SVAR).

7.6 Robustness and limitations

Re-estimation with the March 2020 covariance break yields qualitatively identical conclusions for both specifications. Three limitations warrant statement. First, residual autocorrelation survives at $p = 2$ (adjusted Portmanteau $p < 10^{-12}$); impulse responses are qualitatively unchanged at $p = 3$. Second, the regime-2 variance of the unemployment shock is dominated by the 2020:M03–M05 observations. Third, the break date is treated as known; endogenous Markov-switching estimation à la Lütkepohl and Netšunajev (2017) is a natural extension. Three Cholesky-identified robustness runs, with the option block ordered first and the macro block rotated through three defensible orderings, are reported in the appendix; the option-to-macro variance shares are invariant to the macro-block ordering and the qualitative ranking of channels is preserved.

8 Conclusion

Lenders care about the probability-weighted mass of borrower outcomes below the repayment threshold — a lower-tail object that can vary across distributions with identical variances.

Because index options and corporate debt are priced by the same aggregate state prices, the option-implied distribution of equity returns is a real-time window onto exactly this object, and its higher order moments summarize the priced downside risk that credit-pricing mechanisms transmit into financing costs, production, and prices. This paper develops that mechanism in a deliberately minimal model whose central object is a fragility statistic — repayment obligations relative to downside pledgeable cash flow — and finds its implications in U.S. data. Transmission is state-dependent exactly as an explicit default boundary implies: financing costs are insensitive to priced tail risk until fragility places the downside state beyond that boundary, and recessions proxy for those states. Bringing an observable fragility measure — the NFCI nonfinancial leverage index — into the analysis confirms the reading twice over: it attenuates and sharpens the recession interactions in predictive regressions, and it leaves the local-projection and structural-VAR results essentially unchanged. Tail risk is the dominant real-side channel, contracting production on impact and predicting substantially higher subsequent volatility of unemployment and production in crisis states; the later sign reversals in growth-rate projections are the recovery of depressed levels, as the model's differencing logic implies. Volatility is the dominant nominal-side channel, lowering inflation persistently in crisis states, while the positive tail-risk inflation response is the signature of the financing component, unambiguous for pure repricing in the model. Skewness and kurtosis share a dominant common tail component, justifying a single tail-risk factor. And the option block is predetermined with respect to the macro block, with a state-averaged footprint — 36% of long-run production variance, 18% of inflation variance — that understates its crisis-state importance by construction.

The agenda ahead follows the fragility statistic. Firm-level measures of b_t — leverage, interest coverage, refinancing needs — map to the default-boundary result directly and would carry the cross-sectional predictions: stronger effects for leveraged, externally financed firms, and stronger price responses among constrained firms (Gilchrist et al., 2017). The maturity structure of corporate liabilities would test whether the slow component of the tail channel is strongest where refinancing is slowest. And the term structure of implied moments — three-, six-, and twelve-month tail pricing — would separate transitory from persistent priced risk and discipline the persistence of the tail state directly. If financial markets price the states that matter for credit, the shape of that pricing is macroeconomic data, and this paper is a first accounting of what it contains.

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